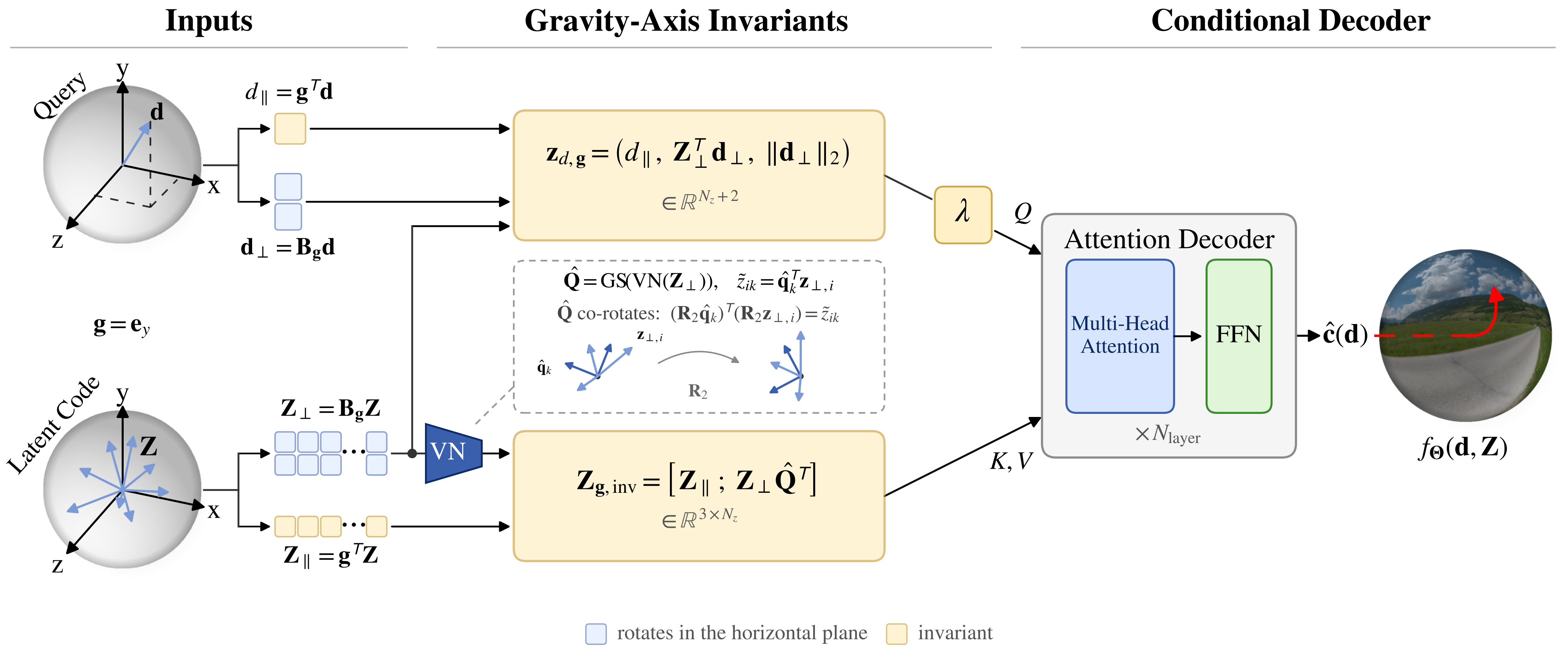
Inputs
Gravity-Axis Invariants
Conditional Decoder
Query
$d_\parallel = \mathbf{g}^T\mathbf{d}$
$\mathbf{d}_\perp = \mathbf{B_g}\mathbf{d}$
$\mathbf{g} = \mathbf{e}_y$
Latent Code
$\mathbf{Z}_\perp = \mathbf{B_g}\mathbf{Z}$
$\mathbf{Z}_\parallel = \mathbf{g}^T\mathbf{Z}$
VN
$\mathbf{z}_{d,\mathbf{g}} = (d_\parallel,\ \mathbf{Z}_\perp^T\mathbf{d}_\perp,\ \|\mathbf{d}_\perp\|_2)$
$\in \mathbb{R}^{N_z+2}$
$\hat{\mathbf{Q}} = \mathrm{GS}(\mathrm{VN}(\mathbf{Z}_\perp)),\quad \tilde{z}_{ik} = \hat{\mathbf{q}}_k^T\mathbf{z}_{\perp,i}$
$\hat{\mathbf{Q}}$ co-rotates: $(\mathbf{R}_2\hat{\mathbf{q}}_k)^T(\mathbf{R}_2\mathbf{z}_{\perp,i}) = \tilde{z}_{ik}$
$\mathbf{z}_{\perp,i}$
$\hat{\mathbf{q}}_k$
$\mathbf{R}_2$
$\mathbf{Z}_{\mathbf{g},\mathrm{inv}} = [\mathbf{Z}_\parallel\ ;\ \mathbf{Z}_\perp\hat{\mathbf{Q}}^T]$
$\in \mathbb{R}^{3\times N_z}$
λ
Q
K, V
Attention Decoder
Multi-Head Attention
FFN
$\times N_{\mathrm{layer}}$
$\hat{\mathbf{c}}(\mathbf{d})$
$f_\Theta(\mathbf{d}, \mathbf{Z})$
rotates in the horizontal plane
invariant